STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-CRITERION-SMOOTHING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3116-3124 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3126-3148, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 23 of queue_examples-defos.json (pb=1121, stmt 3116-3124, proof 3126-3148).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0C3I: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3148–3148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0C3K: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3148–3148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 045U: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3148–3148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02NM: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3148–3148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Set $S = \mathrm{Spec}(k[[t]])$ and $S_n = \mathrm{Spec}(k[t]/(t^n))$. Let $Y \to S$ be a proper, flat morphism of schemes whose special fibre X is Cohen-Macaulay and equidimensional of dimension d. Denote $X_n = Y \times_S S_n$. If for some $n \geq 1$ the dth Fitting ideal of Ω_{X_n/S_n} contains t^{n-1}, then the generic fibre of $Y \to S$ is smooth.*

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By More on Morphisms, Lemma 045U we see that $Y \to S$ is a Cohen-Macaulay morphism. By Morphisms, Lemma 02NM we see that $Y \to S$ has relative dimension d. By Divisors, Lemma 0C3K the dth Fitting ideal $\mathcal{I} \subset \mathcal{O}_Y$ of $\Omega_{Y/S}$ cuts out the singular locus of the morphism $Y \to S$. In other words, $V(\mathcal{I}) \subset Y$ is the closed subset of points where $Y \to S$ is not smooth. By Divisors, Lemma 0C3I formation of this Fitting ideal commutes with base change. By assumption we see that t^{n-1} is a section of $\mathcal{I} + t^n\mathcal{O}_Y$. Thus for every $x \in X = V(t) \subset Y$ we conclude that $t^{n-1} \in \mathcal{I}_x$ where $\mathcal{I}_x$ is the stalk at x. This implies that $V(\mathcal{I}) \subset V(t)$ in an open neighbourhood of X in Y. Since $Y \to S$ is proper, this implies $V(\mathcal{I}) \subset V(t)$ as desired. $\square$